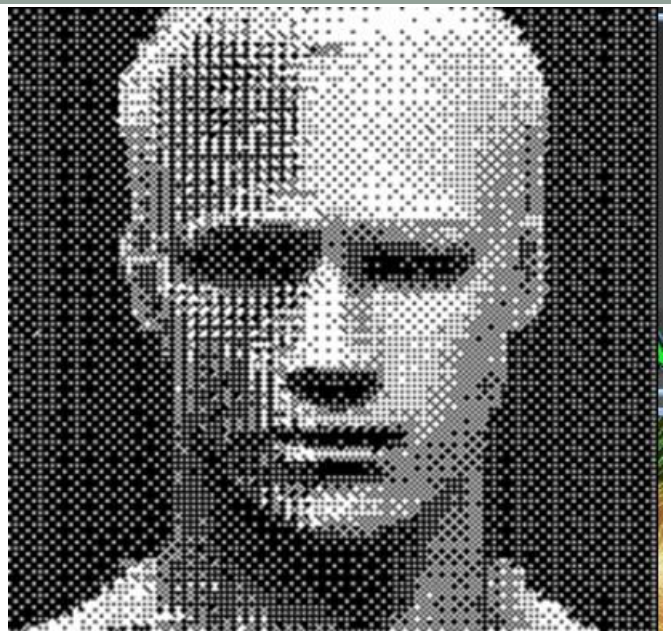


DIGITAL IMAGE PROCESSING

Dr. Manar Donia

CMPn332 - CPN332 & CMP 432

Image enhancement Point processing

Key point processing techniques used in image enhancement are:

- Histogram Equalization
- Contrast Stretching
- **Image Negative**
- **Logarithmic Transformation**
- **Gamma Correction(Power law transformation)**
- **Grey level slicing**
- **Bit plane slicing**

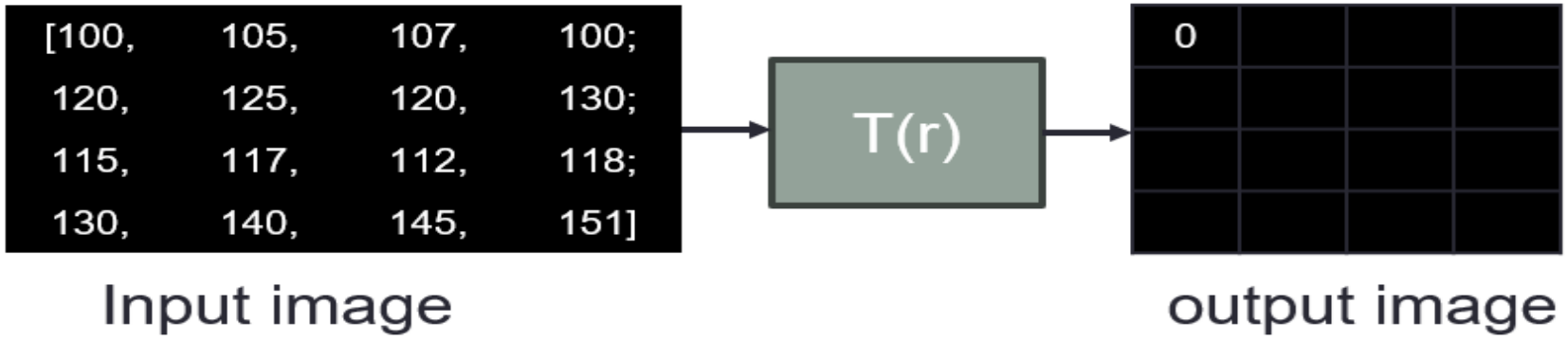
Image enhancement Point processing

The simplest formula for image enhancement technique is:

$$S=T(r)$$

Where:

S: pixel value of output image after processing
T: transformation function
r: pixel value of input image before processing



Histogram equalization

$$S=T(r)$$
$$S= CDF(r)\times(L-1)$$

Image enhancement Point processing

Basic grey level transformation functions used for image enhancement.

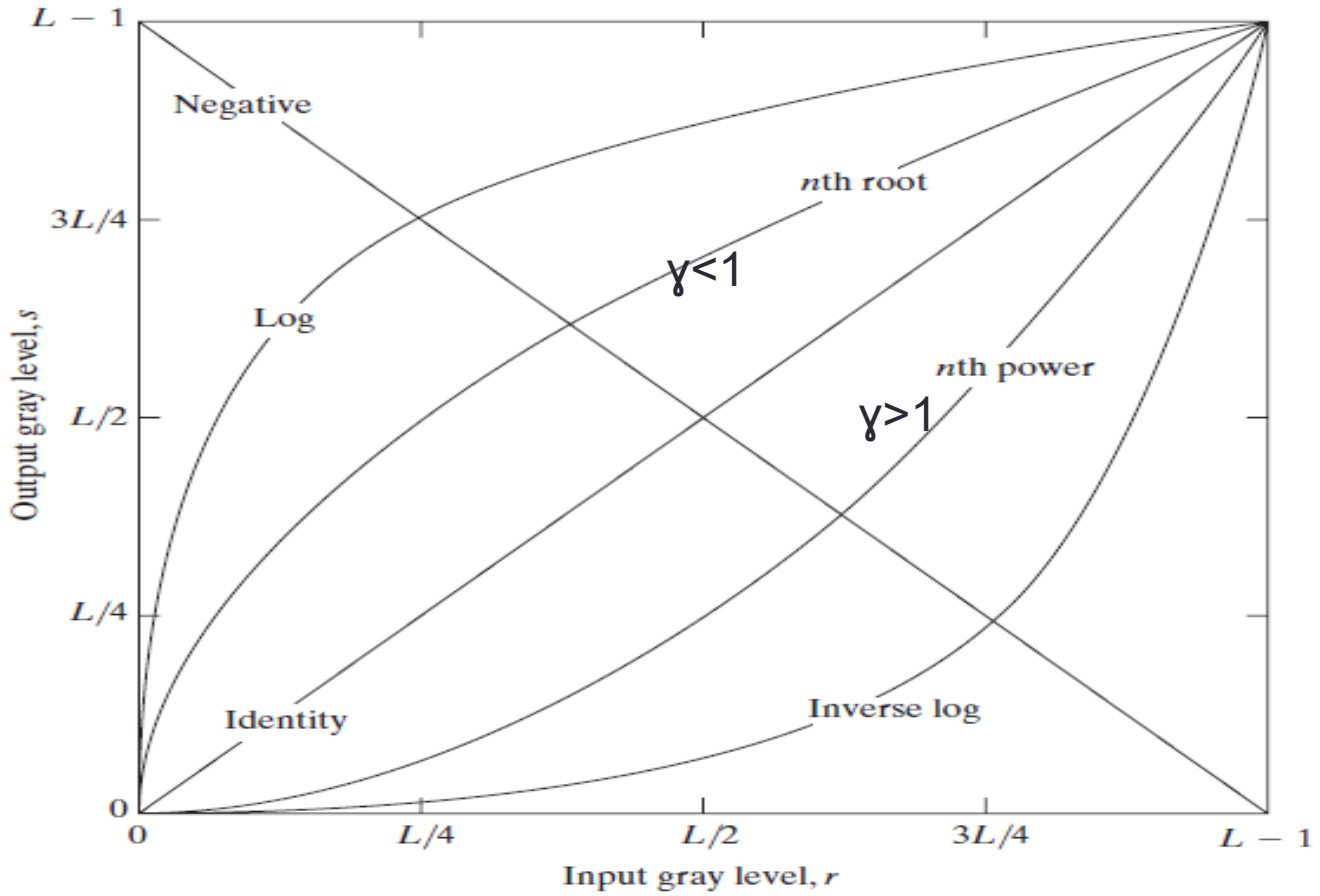


Image Negative

Image Negative are useful for enhancing white or grey detail embedded in dark regions of an image

$$S=T(r)$$

$$S= (L-1)-r$$

Image Negative Algorithm

Input: Image matrix

Start:

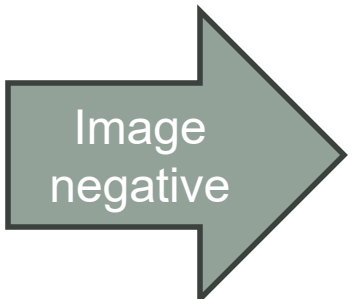
Initialize an output image of the same size as the input image.

For each pixel in the input image matrix do

Calculate the negative pixel value: $s(x, y) = 255 - r(x, y)$

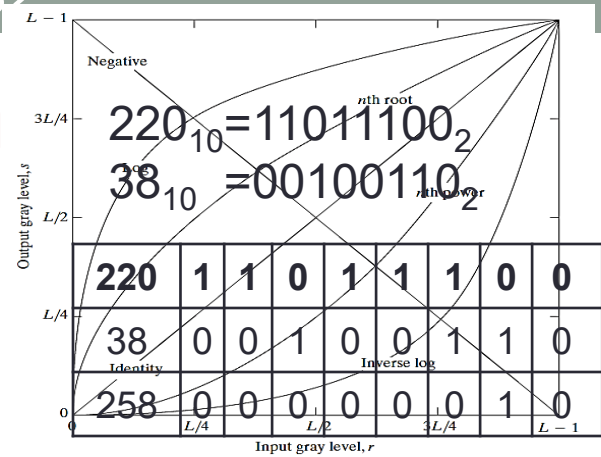
End

[255,	0,	107,	100;
120,	125,	120,	130;
115,	117,	112,	118;
130,	140,	145,	151]



[0, 255,

Logarithmic Transformation



Enhancing image details in darker regions while compressing lighter areas

$$S = c * \log (1 + r)$$

where c= (L - 1)/log(L)

c defined to normalize the resulting image to a valid (visible) range.

For example, intensity level 0:255 log(1)=0 and log(256)=2.408

by using scaling constant c→ $s=\frac{256-1}{\log 256}$*log(256)=255 image

Logarithmic Transformation Algorithm

Input: Image matrix

Start:

- Initialize an output image of the same size as the input image.
- For each pixel in the input image matrix do
  - Increment each pixel value by 1 to avoid log(0)
  - **Apply the Logarithmic Transformation**: $s(x, y) = Log(r(x, y))$
  - normalize the output to ensure that the pixel values fall within the appropriate range by multiplying each S value by C

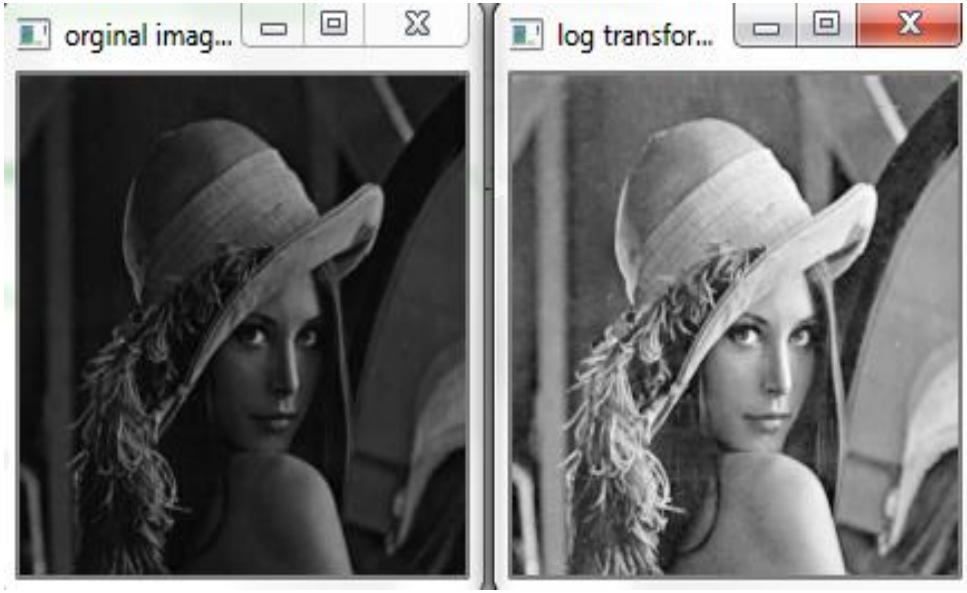
•End

Logarithmic Transformation

[255,	0,	25,	30
120,	125,	50,	10;
115,	117,	112,	118;
130,	140,	145,	151]

Log Transformation

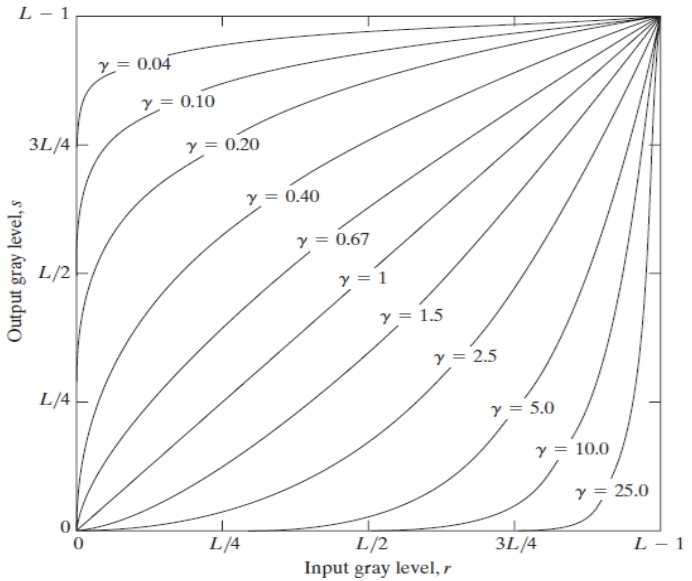
[255,	0,	150,	158;
220	222	181	106
219	219	217	220
224	228	229	232



Gamma Correction(Power law transformation)

- Gamma Correction (power law) depend on the value of γ
  - γ <1 map a narrow range of dark input values into a wider range of output values.
  - γ >1 map a narrow range of bright input values into a wider range of output values.

$$s = c * r^{\gamma}$$ where c=255/ 255^{γ}



Gamma Correction Algorithm

Input: Image matrix

Start:

- Initialize an output image of the same size as the input image.
- For each pixel in the input image matrix do
  - **Apply the Gamma Correction Transformation**: $s(x,y) = r(x,y)^{\gamma}$
  - normalize the output

•End

Gamma Correction(Power law transformation)

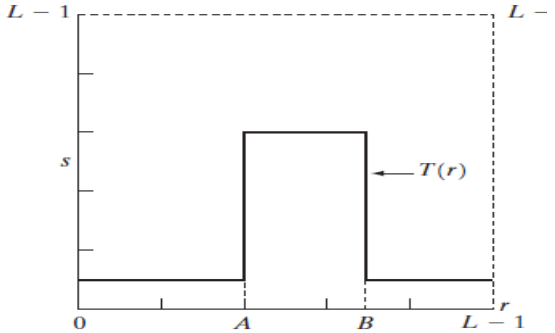


Grey level slicing

Gray level slicing→ It's a way to highlight gray range of interest to a viewer by one of two ways:

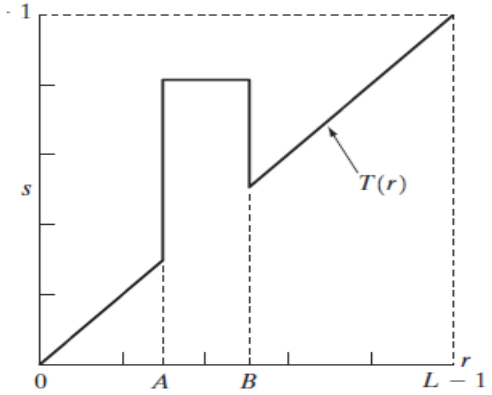
1. Transformation highlights range [A, B] of gray level and reduces all others to a constant level

$$S(x,y)\begin{cases}255 & if\ \ A \le r(x,y) \le B \\ C & otherwise\end{cases}$$



2. Transformation highlights range [A, B] but preserves all other levels

$$S(x,y)\begin{cases}255 & if\ \ A \le r(x,y) \le B \\ r(x,y) & otherwise\end{cases}$$



Grey level slicing

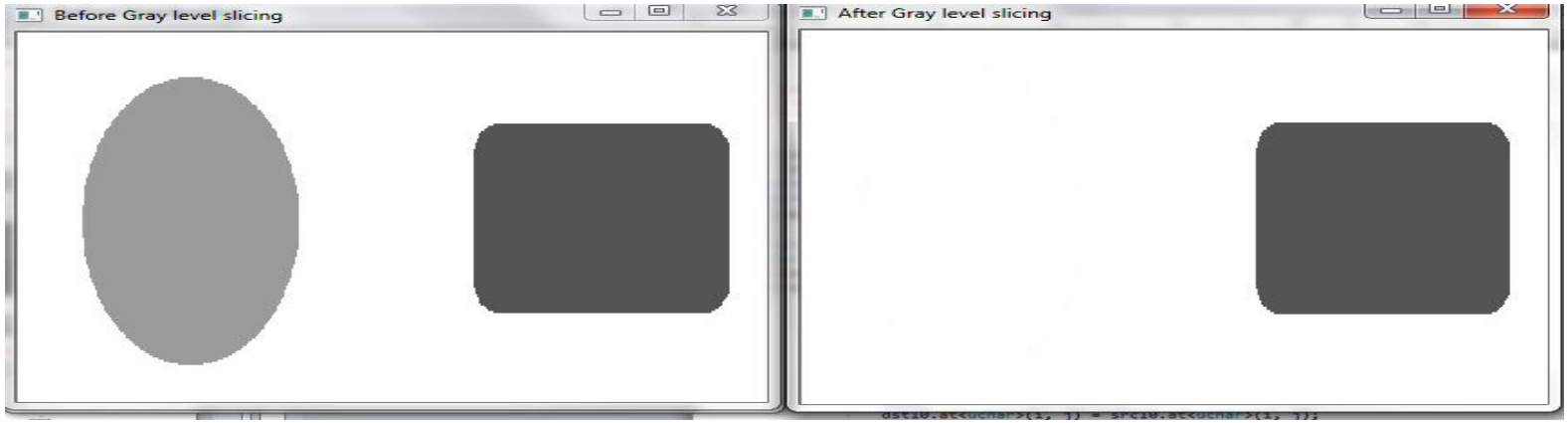
For the following image matrix apply Gray Level Slicing for Range 55 to 100 while reduces all others to a constant level=5

[255,	50,	65,	70
120,	125,	55,	98;
115,	117,	112,	118;
130,	140,	145,	151]

Gray level slicing

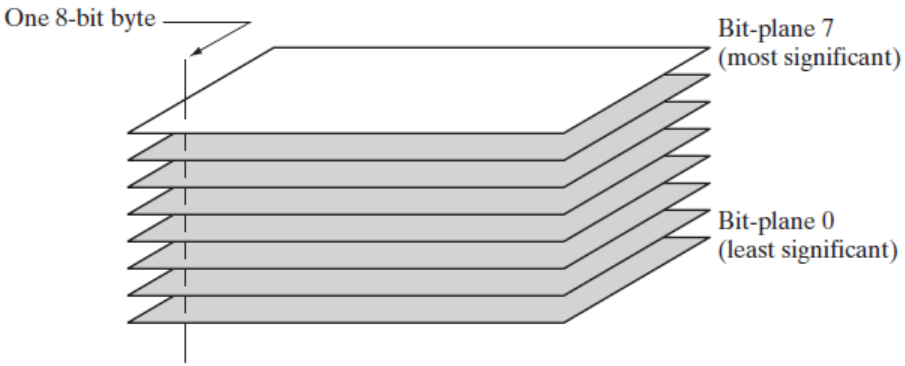
[5,	5,	255,	255;
5	5	255	255
5	5	5	5
5	5	5	5

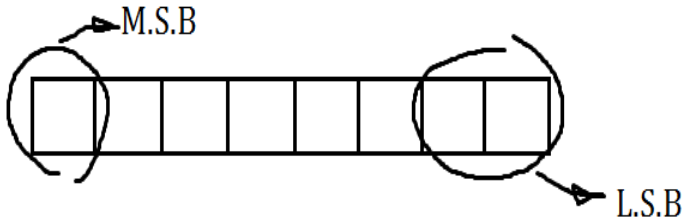
Transformation highlights range [A, B] but preserves all other levels



Bit plane slicing

- Bit plane slicing is a method of representing an image with one or more bits of the byte used for each pixel.
- The main goals of bit plane slicing is:
  - Converting a gray level image to a binary image.
  - Image Compression (Representing an image with fewer bits)





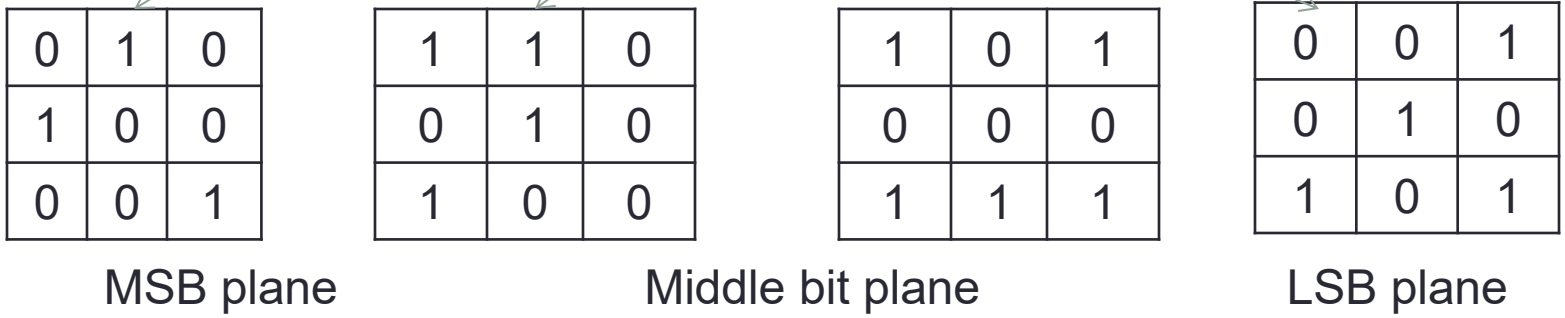
- The binary format for the pixel value 131 is 10000011

Bit plane slicing

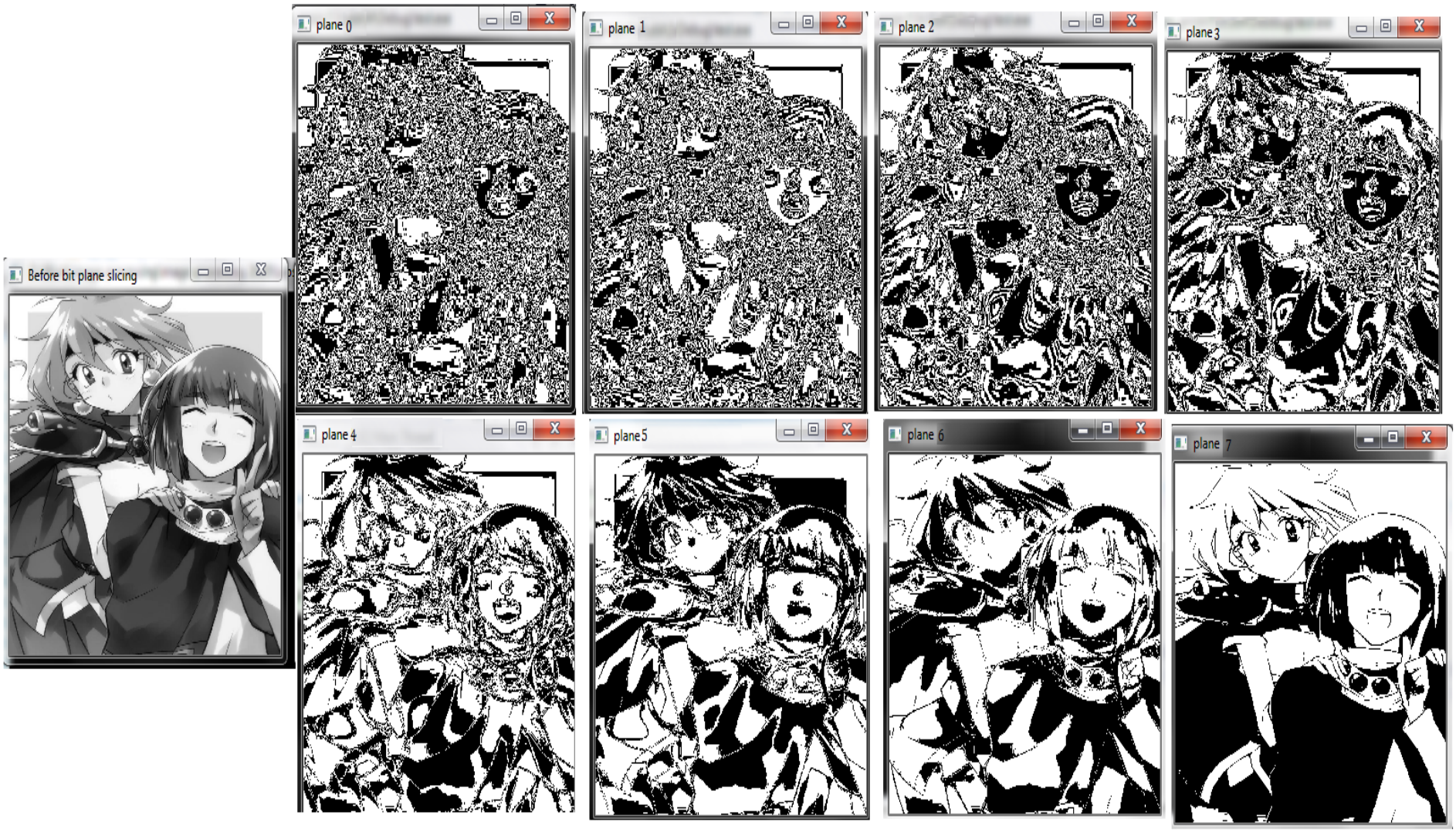
6	12	3
8	5	0
7	2	11

- A 4-bit image has the following intensities
- Find out the bit planes.
- **Answer :**
- We convert the image to binary then separate the bit planes.

0110	1100	0011
1000	0101	0000
0111	0010	1011



Bit plane slicing




Thank you!